

EMERGING TECHNOLOGIES



Technology integration for less commonly taught languages: AI and pedagogical translanguaging

Robert Godwin-Jones, *Virginia Commonwealth University*

Abstract

Less commonly taught languages (LCTLs) have traditionally lagged behind in terms of the availability of learning/teaching materials and appropriate pedagogical models. For many languages, online tools, courses, and digital archives have been developed in recent years that offer opportunities for both formal instruction and self-study. Now the advent of generative AI offers additional possibilities for language practice and the creation of teaching materials. However, AI systems have less robust support for minority languages, especially for Indigenous languages. LCTL teachers and programs also confront the question of whether to adopt the emerging practice of pedagogical translanguaging, which involves integrating learner use of first languages in instruction. For Indigenous communities, the intertwining of culturally significant localities and language, as well as experiences of colonialism, have in some cases led away from online learning options and Western innovations, including AI. The reality for today's learners, however, is that they are living in a world in which AI and multilingualism are fast becoming universal and unavoidable aspects of everyday life across the globe. That fact needs to be considered in navigating technology and pedagogy choices in instructional settings.

Keywords: *Less commonly taught languages; Indigenous cultures; generative AI; pedagogical translanguaging*

Language(s) Learned in This Study: *All languages*

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Introduction

Advances in language technologies have been enormous in recent years. New approaches to natural language processing (NLP), computational linguistics, and related fields have resulted in dramatic improvements in the ability of computers to process human language in both written and spoken formats. That has led to giant strides forward in areas like machine translation, automatic speech recognition, and speech synthesis. At the same time, the use of statistic modeling in generative AI has allowed for the creation of large language models (LLMs), able to interact with humans through language in a variety of ways. Since being released in November 2022, *ChatGPT* has been found to offer extraordinary opportunities for language learning and instruction, as have other generative AI systems (Dizon, 2024; Godwin-Jones et al., 2024a; Kohnke et al., 2023; Poole & Polio, 2023).

For less commonly taught languages (LCLTs), AI would seem to be ushering in a golden age of educational possibilities in its potential to supply teaching and learning materials that have always lagged behind in comparison to commonly taught languages (Villa, 2002; Ward, 2018). Conversation partners may be hard to find for LCTL learners; an AI chatbot supplies an always available, patient interlocutor who can personalize interactions according to learner interests and language abilities (see in this issue: Shin & Choi, 2025; Song & Tang, 2025). However, AI performance is optimized for high-resource languages, especially for English, and is much lower for low-resource languages, which constitute the

vast majority of the thousands of human languages (Atari et al., 2023). Moreover, language output from AI systems for all languages is not linguistically nor culturally authentic, or in many cases, even appropriate (Bender et al., 2021; Godwin-Jones, 2024b). That is particularly the case for LCTLs, and even more for Indigenous languages, which are even further behind in AI performance (McKnight & Shipp, 2024).

That minority languages trail high-resource languages in terms of teaching and learning materials is not a new development. Textbooks (and other pedagogical materials) are less widely available, with many not updated regularly. That holds true for resources in computer-assisted language learning (CALL; Blyth, 2013; Godwin-Jones, 2013; Ward, 2016). Even with the advent of the Internet, instructors of minority languages fell behind in the use of technology compared to more commonly taught languages (Winke et al., 2010). Fortunately, both interest and activity since then have increased considerably, largely through the use of corpus tools, the development of digital archives, and creation of multimodal resources, often shared as open educational resources (OER).

As is the case with all language instruction, organized classes for minority languages have moved online, often using a task-based approach that richly integrates technology resources (Le & Ziegler, 2025; this issue). On the other hand, given the practical constraints of low numbers of learners, virtual exchange (telecollaboration) is less often used than is the case generally today in instructed second language acquisition (SLA; Ward, 2016), although counterexamples show its substantial potential (Wach et al., 2025, this issue). Place-based technologies like digital mapping, augmented reality (AR), and virtual linguistic landscapes offer opportunities to incorporate local environments and to add rich semiotic resources into language learning (Klimanova & Lomicka, 2023). This is a prospect of special potential for Indigenous languages, for which the tight connection of community, place, and language is particularly important (Alberto Mora et al., 2022; Elliott, 2021a).

Another development in language education also has consequences for LCTLs, namely the growing recognition of multilingualism as the reality of most social language use (Ortega, 2013, 2017). The term *translanguaging* has become commonly used in applied linguistics to reference the fact (validated by brain studies) that the languages we speak are not separate in our minds but represent meaning-making resources we can draw on whichever language is in current use (García & Otheguy, 2019; Li, 2018). That has led to the question of what that means for SLA. Advocates argue strongly for integrating pedagogical translanguaging into instruction, i.e. allowing use of students' first language (L1) and any other acquired languages in some learning contexts (Cenoz & Gorter, 2017; Elliott, 2021b).

For instruction in minority languages, there is some understandable hesitation in bringing majority languages or widely taught other languages – most commonly English – into classroom (or synchronous online) settings in which learners have a rare opportunity for intensive use of the target language. That is particularly a concern for Indigenous languages, often endangered, and with a fragile relationship to local majority languages (Mendoza et al., 2024). For those languages, the goals of language revitalization/maintenance and cultural transmission would seem to call for face-to-face, local, immersive teaching environments and away from translanguaging and online learning (Chiblow & Meighan, 2022). On the other hand, for language revitalization efforts, plurilingualism and digital resources may be the only viable approach (Elliott, 2021b).

In this column, we will explore these developments and what they may mean for the future of teaching and learning minority languages. We conclude with advocacy for a balanced and critical approach to integrating new technologies and pedagogies into LCTL instruction. Languages differ widely in terms of social, political, and usage status, and in their relationship to majority languages (see McCarty, 2018), so that local contexts and learning goals will determine the appropriate use of generative AI and of plurilingualism.

Teaching and Learning Materials

Technology has long played a role in research in anthropological linguistics, with analog recordings of speakers used as a resource to preserve the language and to identify its characteristics (Hymes, 1963). The information gathered in that way, and through anthropologists' close contact with speakers of the language, is valuable but does not automatically provide pedagogical resources, as the emphasis has been typically on distinctive linguistic features and cultural practices tied to language use (Bird, 2020; Villa, 2002). Teachers of minority languages have often needed to create their own learning materials or to adapt the available linguistic resources to their teaching context (Elliott, 2021b). In our electronic era, it has become much easier to find and share language learning and teaching resources. That opportunity was initially not widely recognized (Godwin-Jones, 2013). Ward (2018) pointed out that in the [Cambridge Handbook of Endangered languages](#) (2012) only one subsection of 23 chapters discusses the role of technology. It remains true that mainstream journals of CALL contain few articles on minority languages. In fact, applied linguistics in general has not been representative of world languages as a whole, with an overwhelming predominance of studies in English teaching and learning contexts (Ortega, 2017). That applies to CALL studies as well (Buendgens-Kosten, 2020; Sauro, 2016).

Fortunately, in recent years we have seen more studies integrating technology into the teaching and learning of minority languages (Elliott, 2021b). One of the developments has been an increasing interest in creating shared resources (Uebel et al., 2023), accessed or taught online. Corpora have been created, often from collections of social media (*Twitter/X, Facebook, TikTok*) which offer opportunities for finding real-world uses of informal language. These can be used for teaching purposes as well as for the development of language tools like NLP (see Trye et al., 2022 for Māori; Preradović et al., 2019 for Croatian; and Knight et al., 2021 for Welsh). Some Web sites offer substantial resources for encounters with real language use through audio and video. Those are of particular importance for Indigenous languages, for which local learning resources are likely unavailable (Hermes & King, 2013; Olthuis & Gerstenberger, 2019). The [Peskotomuhkati-Wolastoqey Language Portal](#), for example, offers multiple audio recordings and videos of fluent speakers in a variety of everyday situations. Other sites feature podcasts and short form videos (Wagner, 2017), sending an important message to younger speakers that tribal languages fit into their familiar media landscape and can be part of their real lives (Meighan, 2021).

Romero (2021) provided an example of adapting audio resources in K'iche' Maya (spoken in Guatemala) into instructional OERs, with the digital format allowing for exploration of sociocultural variations including regional dialects. The OER materials have been used in formal educational settings but also made available for use by L1 (first language) and heritage learners. Importantly, the learning materials were developed in consultation with Indigenous communities, a necessary step to ensure cultural and linguistic authenticity, as well as to build trust and confidence (Pine et al., 2025; Wiechetek et al., 2024). The instructional materials discussed in Romero (2021), while including lessons on grammar, place a greater emphasis on the development of pragmatic language abilities, crucially important for effective real-world language use. That is also the emphasis in Furniss's (2016) research, which described a project targeting the use of formulaic expressions in Russian. Technology has been shown to be helpful in teaching pragmatics (González-Lloret, 2019; Taguchi, 2022) in that speech acts or other pragmatically informed interactions can be presented dynamically with learner responses to discourse completion tasks and role plays.

The website developed for Russian in Furniss's (2016) work features examples of formulaic sequences drawn from a Russian corpus. That site includes examples of social media use and video excerpts from a multimedia sub corpus. The results indicate that the instructional intervention improved learner awareness and use of targeted routine formulas. The use of examples drawn from corpora provides learners with language use not typically found in textbooks. One learner commented on a learning module on the site which featured *Twitter* (now *X*) examples, "I can't emphasize enough how useful I found Twitter, since it was totally natural, colloquial Russian - which is something that is so hard to come by as part of language learning classes" (Furniss, 2016, p. 49). Online resources present the opportunity to provide many more

examples of real language use than would normally be found in a textbook or typical language classroom. Additionally, students are able, as here, to work with the material interactively and independently, and receive feedback.

Online media offer an ideal delivery mechanism for self-study of LCTLs, as that option fits the profile of many would-be learners of those languages, who tend to be more motivated and older than learners of more commonly taught languages (Ward, 2018). Interestingly, in a study of the learning resources LCTL learners chose to use in self-study, one of the least popular was dedicated language learning apps, such as *Duolingo* (Martin et al., 2024). A learner of Indonesian in that study complained that the app did not provide the “nuances” (p. 86) of how a word might be used in context and commented that “if you were to go to Indonesia and try and base all your language and actions solely off of the app’s language instruction you could end up making a couple of social faux pas” (p. 86). That is the advantage of resources that represent captured, real-world language use, rather than decontextualized dialogues. The connection to actual language use by working with corpora (or engaging in online communities) can be motivating, compared to the artificial and repetitive examples of language in language learning apps (Loewen et al., 2019).

Given that resources for the development of CALL materials for LCTLs are typically limited, it can be beneficial in terms of cost and labor to adapt existing tools and materials (Barcomb & Cardoso, 2024; Ward, 2002). Ward and Genabith (2003) gave an example of modifying a widely used authoring tool at the time (*Hot Potatoes*) to develop a set of learning exercises for Nawat, an endangered Indigenous language of El Salvador. Blyth (2012) described how a Nigerian teacher of French adapted the open textbook *François interactif* to fit a local African setting, a task the teacher assigned partly to her students. The English language game *Cipher* was integrated into an Irish language learning tool (O’Toole et al., 2023). That resulted in a low-cost interactive virtual reality (VR) game, *Cipher: Faoi Gheasa* (Cipher: Under a spell) which most recently has been expanded with AI produced images (Xu et al., 2024). Similarly, Kronenberg and Poole (2022) adapted an existing English language game for learning German.

Gaming has been shown to be a highly motivating approach to language learning (Cornillie et al., 2012; Reinhardt, 2019), including for minority languages (Séror, 2021). In studies of students using a variety of resources for self-study of several different minority languages, it was found that gaming was a learning activity more highly rated than other online activities such as using a language learning app (Mohammed et al., 2021; Martin et al., 2024). Pastushenkov et al. (2025; this issue) found that basing a course for Russian on the gaming platform *Kahoot* was pedagogically effective and motivating for the majority of learners. Poole et al. (2025, this issue) used gaming both as an instructional and assessment tool for beginning Chinese learners.

The COVID pandemic led to all language classes going online. In fact, that movement was already underway. The *STARTALK* online courses, for example, had pioneered online courses for LCTLs since 2007 (Nier et al., 2009). Given that enrollment numbers are likely to be low in most minority language classes, online formats offer the opportunity to combine individuals and groups of learners into a single cohort. That practice has been used for some time among universities in the United States (Galvin et al., 2023) and across Europe (*European Language Equality Network*; see also Steingrímsson et al., 2024). Online courses bring together students from different localities, and likely with different linguistic backgrounds. That might include heritage speakers as well as absolute beginners, which can pose challenges pedagogically (see Le & Ziegler, 2025, this issue).

In online course design, as in developing CALL resources, as discussed above, it is advantageous to draw on existing models and frameworks in designing and developing online courses for LCTLs. Le and Ziegler (2025, this issue), for example, apply a task-based online curriculum (Chong & Reinders, 2020; Ziegler 2016) to a beginner's course in Vietnamese. That study showed the benefit of integrating tasks that had real world connections, as often stressed in the discussion of technology-informed task-based learning (González-Lloret, 2019; González-Lloret & Ortega, 2018). Other studies examining technology

integrating task-based learning have likewise found it beneficial to organize tasks that are meaning-focused and contextualized for authentic communication (see Kim et al., 2022 for Korean; Lai et al., 2011, and Nielson, 2014 for Chinese).

Minority Languages and SLA Theory

Despite the examples discussed above of projects integrating technology into the teaching and learning of minority languages, it is still true, as it was a dozen years ago, that the vast majority of published studies in CALL represent majority languages, most often English (Godwin-Jones, 2013; O'Toole et al., 2023). The lack of representativeness in applied linguistics of the majority of the world languages is clearly problematic for assertions of generalizability of study results for language learning pedagogy as a whole (Ortega, 2005; Sato & Loewen, 2022). That applies to CALL studies as well (Ortega, 2017). Cenoz and Gorter (2024) pointed out that including a wider range of learning and teaching contexts, in particular minority languages, would enrich research in SLA. That inclusion of minority languages would, for example, serve to demonstrate the artificiality of accepted binaries in applied linguistics, such as “foreign language” versus “second language” learning. Cenoz and Gorter (2024) argued that the study of minority languages typically falls in between the two. Minority languages are typically learned in the country where they are spoken, but because of the presence of a majority language, most learners have less access to using the studied language in everyday contexts.

The assumed goals of instructed SLA may gain a new perspective by considering the different expectations which instruction in minority languages may have. Instead of aiming for four-skills fluency, some instructors and programs may have goals that involve learning enough of the language to satisfy linguistic parameters of representing authentic Indigenous identities (Bird, 2020; Elliott, 2021b). For language revitalization projects, Western pedagogies may not meet the needs of local communities where goals might center on intergenerational language transmission and the promotion of language use in particular social domains (Meighan, 2021). Learners seeking to connect with ancestral languages may be less concerned with gaining the ability to use the language transactionally in everyday life but rather, as a government survey of Australians demonstrated (Marmion et al., 2014), to strengthen the connection to the culture and build a sense of identity and personal well-being. Indeed, Aboriginal Elders, as may be true in other Indigenous communities, are likely to place a higher value on the transmission of cultural knowledge and traditions than on obtaining speaking ability (Bird, 2020).

Integrating pedagogies used for minority languages into SLA theory illuminates multilingual realities. That applies to Content and Language Integrated Learning (CLIL), which has a long track record of recognizing, instituting, and promoting plurilingualism (Cenoz, 2015). A description that has arisen in studies of minority languages, namely *new speakers* (O'Rourke et al., 2015; McCarty, 2018), shows the potential for enriching SLA theory. The term points to learners with no home or community exposure to a language, but who acquire it as an adult through bilingual or revitalization projects (Pujolar, 2022). That terminology is helpful in moving away from the problematic and divisive dichotomy of “native” versus “non-native” speaker (Dewaele et al., 2022). The *new speaker* moniker, similar to *LX*, (i.e., additional language) “circumvents the negative ideological connotations that have become naturalized in the term ‘native’ and ‘second’ in scholarly usage” (Dewaele et al., 2022, p. 36).

The use of *L1 speaker*, along with *new* or *LX* speakers, index the chronological acquisition of languages and imply a status of equality and neutrality among languages. Calling for equal respect among languages should not be misunderstood as an assertion of equality in the positioning of languages in multilingual contexts, in which majority languages (predominantly English) inevitably exert greater power and prestige, a reality to take into account in plurilingual pedagogies (García, 2019; Mendoza et al., 2024). Indeed, one of the benefits of pedagogical translanguaging, as discussed below, is to raise critical consciousness about the power differential between minority and dominant languages (Muñoz-Basols et al., 2023).

That positioning among languages towards equality is of particular importance for Indigenous languages for which there is a long history of prejudicial social treatment, political persecution, and academic neglect (Nyaaba et al., 2024). Greater consideration of Indigenous languages in SLA could have the beneficial byproduct of bringing into theory and practice views on communities and pedagogies from the Global South (Canagarajah, 2021). McKinley (2022) suggested, for example, that a more globalized approach to teaching writing might lead to more acceptance of “creative imitation” (p. 4) in LX, generally seen as an unacceptable approach to essay writing in second language writing. Communal values and practices associated with First Nations, such as acceptance of ambiguity and difference, as well as acknowledgement of the tight bond between humans and the environment, can be helpful in broadening the scope of theory (Canagarajah, 2021), as well as to conceptualize the agentic role of AI in human society (Godwin-Jones, 2024a).

Plurilingualism and LCTLs

In addition to the non-representation of most of the world languages in academic scholarship, representativeness in SLA studies is also challenged by the demographics of the studied groups, which overwhelmingly consist of tertiary level learners, typically in well-funded settings (Byrnes, 2020). Of particular relevance in the current world of mass demographic displacement and immigration is language learning in the context of refugees and immigrants. Yilmaz (2024) studied the “linguistic precariat” (p. 1) of refugees from Africa living temporarily in Greece, waiting to gain entrance to northern Europe via human smugglers. The refugees’ experience represents a complex linguistic environment, communicating among fellow refugees and with family back home in their L1, while being invested in learning both English and Greek, and possibly German or Swedish. In the process, they make rich use of mobile technology for communicating and learning (see Deumert, 2014; Godwin-Jones, 2017; Kukulska-Hulme et al., 2015). Examining minority language use inevitably brings into consideration additional languages, most often English, as majority or lingua franca languages. Multilingualism is a reality for most of the world’s population and has become ubiquitous in electronic media (Godwin-Jones, 2018, 2023b). The everyday multilingual experience of most humans points to the outdated monolingual orientation of much of the practice of instructed SLA (Ortega, 2013, 2017).

Studies of LCTLs demonstrate the problematic dichotomies of LX user versus LX learner as well as of native versus non-native speakers. As Cenoz (2019) discussed in the context of Basque, multilingualism is “more complex than a simple dichotomy” (p. 133), as speakers of that language in Basque communities (in Spain and France) have a wide range of levels of proficiency. In that way, abilities lie somewhere along a continuum. Speakers of Basque (and other minority languages) use the language in different ways in different contexts, with the majority language always in the background (or foreground). In that way, boundaries between languages are “soft” (Cenoz & Gorter, 2017, p. 74), not fixed and solid. The plurilingual reality of learners means that they are not “empty vessels” (Cenoz & Gorter, 2020, p. 1) to be transformed into monolingual speakers of the target language. Rather than seeing learners as incomplete LX users on their way to becoming fluent speakers, we should instead conceive of them as “skilled communicators” (Cenoz & Gorter, 2017, p. 75) who are able to draw on all the semiotic resources at their disposal, no matter the language spoken at the time.

Enabling learners to take full advantage of their semiotic resources is likely more feasible in online environments than it is in the classroom. In instructed environments, the dictates of fixed, standard curricula, official local or national standards, the need to prepare students for high stakes exams, and entrenched beliefs on optimal teaching practices may well minimize students’ use of their L1. That can be particularly the case for the instruction in Indigenous and threatened languages, for which immersion may be seen as a necessity in revitalization efforts (Chiblow & Meighan, 2022; Hermes, 2014). Online students will inevitably encounter code mixing, as studies have shown the frequent plurilingualism in social media including *Facebook* and short messaging services (Lee, 2016; Ortega, 2017). Speakers of Indigenous languages in recent years have been shown to be avid users of social media, often integrating

multiple languages (see Galla, 2016; Meighan, 2021). Studies have demonstrated how LX learners online are able to explore and express their multilingual identities, richly tapping into their personal store of languages (Godwin-Jones, 2018; Røyneland & Cutler, 2018). Oliver and Nguyen (2017) discussed how Aboriginal students mix languages in *Facebook* posts, leading to group solidarity, humorous language play, and improvement in standard English skills.

Pedagogical Translanguaging

The continuous presence of the resources in one's brain (and body) of all the languages we know has been characterized as *translanguaging*. While that concept has been generally accepted in applied linguistics, its manifestation in instructed SLA, in the form of pedagogical translanguaging, has been less widely implemented. That is particularly the case for minority languages, where there is a perceived need for a delicate balance between immersion and multilingualism in instruction. Some inclusion of students' L1 (and of any other known languages) in SLA "has been supported by an increasing number of theoretical position statements" (Brown, 2023, p. 1639), some older (Atkinson, 1987; Cook, 2001) and others more recent (Ortega, 2017; Byrnes, 2020). Despite those findings, "professional practice in this domain seems quite resistant to change" (Brown, 2023, p. 1639). That is perhaps not surprising in that mainstream SLA theory, exemplified in the Communicative Approach (Savignon, 1991), has long advocated maximizing the target language and minimizing L1 in the classroom. L1 use is closely tied to outmoded teaching approaches like grammar-translation. American Council on the Teaching of Foreign Languages (ACTFL) guidelines in the United States have called for [90% use of the target language](#) in classroom instruction. In fact, the need to avoid the L1 in SLA instruction appears to be deeply entrenched in both teachers and learners (Byrnes, 2020), a viewpoint often echoed in official pedagogical guidelines and supervisory agencies (Brown, 2023). Teachers engaging in translanguaging practice may experience unease, even guilt, and could face professional penalties (Byrnes, 2020).

Despite the concerns and controversy, multilingual practices have found their way into instructed SLA, as studies attest (see special issues on pedagogical translanguaging in [System](#), 2020 and in [Pedagogies](#), 2022). Most of those have focused on teaching English, with few exploring translanguaging with minority languages. Most studies are descriptive rather than empirical, so it is difficult to extract clear evidence of improved learning outcomes through translanguaging (Byrnes, 2020). It is also the case that the process of integrating translanguaging may change expected (and measured) outcomes, with greater emphasis on assessing students' overall language abilities and effectiveness in using all available semiotic resources to communicate (García & Otheguy, 2019). Moreover, the contexts in which an additional language is taught vary so greatly—and the different potential uses of the L1 are so mixed—that generalization is on shaky ground. However, studies are helpful in illuminating possible routes to beneficial integration of L1s in SLA, as long as it is understood that they serve as examples of what has been possible in a particular teaching context (see Byrnes, 2020). Those studies which deal with target languages other than English, as summarized in Brown (2023), demonstrate that integration of the L1 can play a mediating and scaffolding role, enhancing language play and learner motivation, and increasing the effectiveness of student interactions in collaborative activities. Pedagogical translanguaging has been shown to foster both learning language learning and content development (Byrnes, 2020; Cenoz & Gorter, 2017).

Brown (2023) is one of the few empirical studies that has compared monolingual versus multilingual instruction for a minority language. It was conducted over a ten-week period for novice learners of Arabic and French and found that groups experiencing multilingual instruction outperformed those in monolingual instruction. That study involved exclusively face-to-face instruction. Indeed, learners were instructed to spend no more than one hour a day studying the language outside of class, so as to achieve some uniformity in student learning activities. In that way, the study is of interest in demonstrating the effectiveness of a multilingual approach in a controlled classroom setting but is likely not representative of how most language learners, including those of minority languages, go about learning. That process likely involves considerable extramural time studying and using online materials. In fact, studies in recent

years have demonstrated the widespread use of online sources for language learning, including social media, affinity groups, and streaming media (see Sockett, 2014; Sundqvist & Sylvén, 2016). The vast majority of those studies, however, were done on learners of English, for which abundant high quality and popular videos and music are available.

Students can and do use online materials for LCTLs and Indigenous language learning, including dedicated apps such as *Duolingo*, which supports over 100 languages. Streaming services have increasingly supported more languages. *Netflix*, for example, is available in 190 countries and supports 60 languages. *YouTube* is offered in 80 languages in over 100 countries. Those resources can be integrated into instructional settings to provide students with opportunities for language practice and cultural learning. A framework for enabling that is the use of *bridging activities*, which involves learners seeking out target language resources online (artifacts or communities) which they experience outside of class, and which subsequently are integrated into classroom discussions or other organized learning activities (Thorne & Reinhardt, 2008; Yeh & Mitric, 2023). The possibility of integrating bridging activities depends of course on the availability of target language resources online, which will vary. In most cases, some forms of social media will include target language use, although not exclusively. For Korean learners in the United States, Reinhardt and Ryu (2013) discussed a project in which students participated in Korean language *Facebook* groups in order to practice the use of honorifics and other pragmatic language usage. The researchers found that the experience contributed to the development of socio-pragmatic awareness. That included some students engaging in language play by flouting pragmatic conventions in order to gauge reactions. Other studies have dealt with different social media, including *Instagram* (Darvin, 2023a) and *Reddit* (York, 2023), but have focused mostly on learners of majority languages (see, however, Hwang et al., 2024; Wagner, 2017).

An illustrative study that looks at the use of technology in a plurilingual setting examines the use of *Google Translate* by a developing bilingual elementary school student (Vogel et al., 2018). The study discusses the multiple ways that a young L1 Chinese speaker moves between languages and online tools to build a personal repertoire of communication resources that include verbal language but also nonverbals and pictures. In that way, translanguaging practices extend to multimodal resources, and potentially to AI as well. In the process learner agency and identity formation are enabled. Thus, pedagogical translanguaging can not only play a mediating and scaffolding linguistic role for learners but can also transform their personal lives (Alberto Mora et al., 2022; Tzirides, 2022).

Indigenous Languages and Place-based Learning

The reality for LCTLs is that it will likely be considerably more difficult to find online affinity groups in which the target language is as prominently used on a platform as is English in most social media (Ward, 2016). Exceptions exist, such as *Facebook*-like platforms for Russia (*Vkontakte*) or the various Chinese social platforms, but they are few and less diverse in terms of content, user profiles, and acceptability of outside participants. That will likely be even more the case for Indigenous languages. In some Indigenous contexts, there may not be great interest in leveraging learning resources outside of the local community. Indigenous language revitalization and maintenance programs are typically tightly integrated into local cultural traditions and in fact tied to the community and its land. Canagarajah (2021) emphasized “land-based pedagogical and communicative practices” (p. 579) in the teaching of Indigenous languages, a perspective echoed by Henne-Ochoa et al. (2020): “Indigenous understandings of language are intertwined with Indigenous concepts of land, identity, and thought, and as such, cannot be successfully compartmentalized and transmitted” (p. 482).

Chiblow and Meighan (2022) commented that when it comes to Indigenous languages, “language is land, land is language.” That can take the literal form of learning a language in place, walking and talking in natural settings, such as a walk in the woods, as a basis for an Ojibwe language learning lesson (Engman & Hermes, 2021). Place-based language learning has gained interest in SLA in recent years, spurred on by interest in semiotics, GPS tracking, linguistic landscapes, and AR (Klimanova & Lomicka, 2023). That

extends to minority languages as well, as shown in the case of linguistic landscapes in learning Basque (Cenoz & Gorter, 2017) or multilingualism in AR (Godwin-Jones, 2023a). Digital maps can offer a contrastive representation of colonized lands, as in the [map](#) of Canada created for the “Come Home to Indigenous Places” project (Pearce, 2017), which shows the location of Indigenous tribes (in the Inuit and Métis languages) and leaves out modern urban centers all together. AR represents an exceptional opportunity for connecting language, place, and culture. Studies have shown how AR projects can be integrated into Indigenous language and culture learning, with place-based stories and historical accounts overlaid as text, image, or audio (Cooper & Kruglikova, 2019; Marques et al., 2019; Zapata-Paulini et al., 2019).

The emphasis in Indigenous language learning on local place, cultural learning, and traditional life experiences, is often reflected in pedagogical approaches, leading in some cases away from translanguaging. In the teaching of Australian aboriginal languages, Chiblow and Meighan (2022) stressed the importance of immersive environments in instruction, “in order to avoid reducing Indigenous worldviews and ideas into ill-fitting English words, binaries, or colonial mindsets” (p. 209). The importance of transmitting traditional beliefs, practices, and values is widely shared by Indigenous educators (Canagarajah, 2021; Cenoz & Gorter, 2017; Whyte, 2013), for example, ways of being and knowing that are representative of Aboriginal traditional knowledge (Cumming-Potvin et al., 2022). That knowledge is passed on by Elders but is also embedded in language so that “language learning and transmission take place on the land, in the home community, and in the language” (Chiblow & Meighan, 2022, p. 209). That perspective can be understood and should be acknowledged and valued given the violence perpetrated by colonial powers in an attempt to forcibly suppress Indigenous languages and cultures. It also, however, has the practical consequence of making it problematic for anyone not part of the local community to learn an Indigenous language. It precludes in some contexts placing value on making online learning materials available. The skepticism over the use of technology in learning Indigenous languages is shown in a comment from a young Aboriginal in an open forum, “We need to prioritise ‘on land’ over ‘online’... you don’t revive a language with an app but with people reconnecting with people and country” (cited in Bird, 2020, p. 3504). The comment was greeted with applause, demonstrating that this was a widely held view (Bird, 2020).

An insistence on the priority for Indigenous languages of local immersive learning environments also eliminates the option of translanguaging, which from that perspective “may exert detrimental influence on the maintenance of minority languages” (Bynres, 2020, p. 3). An alternative that offers a possible middle ground is the practice of responsive “sustainable translanguaging,” designed to protect and maintain endangered languages and which takes into account that the study of minority languages will look different in different settings (Cenoz & Gorter, 2017, p. 901). In a study centering on the revitalization efforts in support of the te reo Māori and Samoan languages in New Zealand, the leaders of the project made sure that rich opportunities for Indigenous language use were present, while also allowing for the use of English on an as-needed basis (Seals & Olsen-Reeder, 2020). That included allowing for both spontaneous (multilingual usage naturally occurring) and intentional (instructional) translanguaging (see Cenoz, 2019), with the goal of building enhanced semiotic resources that are multilingual. That process resulted in a set of “pedagogical rules” (Seals & Olsen-Reeder, 2020, p. 5) developed via close contact with community members through public workshops. Those rules guided both the writing of learning materials and instructional interactions. The public workshops and the transparently developed and shared rules were designed to address community concerns and to share evidence of successful language learning in multilingual settings.

The New Zealand project targeted preschool children and was entirely face-to-face. Those same principles of sustainable translanguaging could be applied to CALL resources such as the creation of online materials, setting up collaborative opportunities and tools, and establishing and monitoring communication channels. That process would need to include consultation with community members to ensure that cultural values are integrated, and that trust has been established. Safe and secure spaces for both single language use and multilingual practice—each with appropriate guidelines—would need to be

available. That might take the form of setting up particular areas within social media reserved for Indigenous language exchange. The reality for most learners is that they will live in a world—both online and in person—that is multilingual. A vital goal for all instructed SLA should be preparation for participating fully in that environment. That includes Indigenous communities, which, like the Māori community described in Seals and Olsen-Reeder (2020), today may have few if any monolingual speakers of the language, even if that was the case historically.

A model for establishing responsive translanguaging online while emphasizing Indigenous language learning is provided by the [Oceani3](#) initiative, established between universities in France and Spain, along with a Basque nonprofit, featuring the use of French, Spanish, and Basque. The discussion forums and online meetings have clearly stated guidelines for language use, with a privileged position for Basque as a minority language. That project is content-based, focused on water pollution, but with careful attention paid to equitable multilingualism (Ortega, 2017). Another example of a multilingual framework applied to an electronic medium is a programming course in California featuring the use of the coding learning platform [Scratch](#) (Jacob & Warschauer, 2023)—that and other studies have shown that a multilingual setting aids content learning and also enhances learners’ self-respect, confidence building, and identity formation that validates learners’ linguistic backgrounds (Jacob et al., 2022; Nasir & Vakil, 2017). That is a hoped-for byproduct of CLIL instructional models. Key to the success of CLIL, according to Fortanet-Gómez (2013), is what is also crucial in sustainable translanguaging, namely that all stakeholders be on board and that the implementation not be linked to a single course or teacher but represent a “systemic embeddedness of change” (Byrnes, 2020, p. 8).

Involving local community members meaningfully in the development of technology integration into Indigenous language learning projects serves to counteract “negative and Western or colonial imposed stereotypes” (Meighan, 2021, p. 403), which view Indigenous people as being antiquated and rigidly place-bound, perpetuating a museum-like identity for those communities (Bird, 2020). Projects which incorporate leading edge technologies like AI and VR are particularly effective in transmitting an alternative perspective on Indigenous communities. That image is sent not just to the outside world, but importantly also to younger members of the local communities, demonstrating that the Indigenous cultures (and languages) have relevance in today’s world. [Te Hiku Media](#), headquartered in Kaitia, New Zealand, first created a digital archive of Māori speakers, then used that resource to build NLP capabilities and has made most of its voice technology tools available as APIs through its digital language platform, [Papa Reo](#) (Hao, 2022). VR projects can be of particular interest to Indigenous communities in that they use sound and image technologies to create a mirror world of local places of cultural importance. [Aboriginal Territories in Cyberspace](#) is a research and production network creating virtual worlds and video games (Meighan, 2021). Through the use of geolocation, AR, and VR, [OurWorlds](#) is working to create immersive environments that portray Native American stories and cultures. As in the development of any technology project involving Indigenous communities, it is vital with AR and VR to involve local representatives from the earliest stages of development to ensure both authenticity and acceptance (Moshagen et al., 2024) and to show that the project does not represent “digital neocolonialism” (Nyaaba et al., 2024, p. 1).

Opportunities and Challenges of AI

If immersive technologies and pedagogical translanguaging are both potentially suspect for members of Indigenous communities, that is likely to be all the more the case for the possible use of AI in language education. That skepticism is in this case well founded. It turns out that large language models are not so “large” as to include the bulk of the world’s languages, as 93% of the existent languages are not represented in mainstream AI systems (Wong, 2024). That includes large numbers of Indigenous languages as well as other minority languages (and dialects). The number of languages officially supported by AI platforms is deceptive (80 for ChatGPT, 40 for Gemini) as the performance in languages is quite different if they represent a large-resource language versus low-resource language. Studies have

shown significantly degraded performance in low-resource languages (Lee & Wang, 2023; Su & Goslar, 2023). Standard American English is the default in generative AI systems (Smith et al., 2024). In fact, in parsing and outputting text in a particular language, the data is often first filtered through English (Stokel-Walker, 2024).

Cultural authenticity is another major issue. The training data used by AI systems is taken from digitized text collections (mainly on the Internet) which are heavily oriented towards English language culture and Western values (Atari et al., 2023). That orientation has been described as WEIRD (Western, Educated, Industrialized, Rich, Democratic) (Henrich et al., 2010) and does not in fact reflect the majority of the world's cultures. That has been demonstrated in studies, such as those of Asian (Wang et al., 2023) and East European (Masoud et al., 2023) cultural contexts. Naous et al. (2023) looked at cultural sensitivity in Middle Eastern settings and in Arabic language interactions with AI. The study found that *ChatGPT*, upon being asked to supply suggested activities after an Islamic prayer session, recommended going out to a bar for a drink, clearly not appropriate in that cultural setting. That study also found that in storytelling, characters with Middle Eastern names were described in much less positive terms than those with Western names. Nyaaba et al. (2024) provided multiple concrete examples (in texts and images) of cultural distortions produced by AI, showing discrimination against non-Western people and places.

The situation for Indigenous languages is even worse. When asked to provide output in Fon (spoken in Benin and neighboring countries), *ChatGPT* replied that it was not possible to do so, as that was a fictional language (Wong, 2024). Therefore, it is not surprising that members of Indigenous communities have very ambivalent feelings towards their languages being incorporated into AI learning data. On the one hand, exclusion means that speakers of the language will not have the same benefits as speakers of majority languages, such as engaging in queries/search/chat in their preferred language or using AI in that language to generate text output, translations, or images. Not having full access to AI in today's world can also have consequences in terms of academic performance and professional advancement. On the other hand, if an Indigenous language were to be included in AI, the limited corpus of available (digital) texts will likely result in inferior performance. That can be seen in the experience of learners of Maltese, a Semitic language indigenous to Malta, who found that *ChatGPT* was incapable of communicating effectively in that language (Żammit, 2024).

It is also the case that colonial and post-colonial mistreatment and neglect of Indigenous peoples and their cultures have led to distrust and suspicion of government and education authorities (Elliott, 2021b). That may extend as well to researchers and AI engineers, making collaboration and data collection difficult (Bird, 2020). Ward (2018) pointed out that the development of CALL materials for an Indigenous language may be “perceived as a threat to traditional pedagogical practices or intergenerational authority relations” (p. 125). That parallels resistance in Indigenous communities to translanguaging and to other examples of technology integration. That possible built-in resistance makes it all the more important that AI projects do not imitate the experience in field linguistics of the “parachute linguist” (Elliott, 2021a, p. 446), extracting data but not contributing to the welfare of the Indigenous community. Collaborative relationships are needed both with commercial AI companies and with any academic researchers involved.

Cooperation with AI companies is difficult (Moshagen et al., 2024) as they are commercial entities whose purpose is to make a profit. If a given language has a relatively few number of speakers, there will be less demand for AI functionality in that language, resulting in profit-driven decisions not to invest the substantial sum involved in adding that language to its system. It is also the case that AI companies are not transparent about their data sources. That fact, along with the resistance in some minority language communities to cooperate with outsiders, has led some researchers in AI-underserved language communities to initiate projects to supplement existing AI systems or to create their own AI. That is a difficult and expensive enterprise, requiring local language expertise, technical know-how, and sufficient human and monetary resources.

For Indigenous languages there are numerous obstacles in the way of creating their own NLP tools or AI systems. Many languages have no writing system or have only recently developed written forms, for which there is scant data (Moshagen et al., 2024). Others may not have standardized the written language so that there is great variation in spelling and orthography (Wiecheteck et al., 2024). Languages (especially in India) may have multiple writing systems (Panigrahi, 2023). Since generative AI depends on substantial text input, small amounts of written data are insufficient for the development of full-blown language models. In addition, digitized texts available for some languages may be produced by machine translation or by language learners, thus not representing reliable, idiomatic, and grammatically accurate language. Possible workarounds include using rule-based AI rather than machine learning (see Wiecheteck et al., 2024) or using transfer learning techniques to add indigenous language capabilities to already existing LLM's (Moshagen et al., 2024; Pinhanez et al., 2024).

It is encouraging that despite those obstacles a number of initiatives have been launched to improve AI performance for minority languages. The Cheetah project, for example, is developing an “Afrocentric NLP” (Adebara et al., 2024) to support 517 low-resource African languages. [Masakhane](#) is a grassroots organization whose mission is to strengthen and spur NLP research “in African languages, for Africans, by Africans.” Meta’s [No Language Left Behind](#) project (NLLB Team, 2022) is developing open-source models for translations directly between 200 languages—including low-resource languages like Luganda and Urdu, without having to go through English. The [Aya project](#) is a global initiative involving researchers in 119 countries seeking to develop a multilingual AI for 101 languages as an open resource. That system features human-curated annotations from fluent speakers in many languages. In North American, the [Indigenous Protocol and Artificial Intelligence Working Group](#) is working to develop new conceptual and practical approaches to building AI systems from an Indigenous perspective.

In addition to locally based projects involving written datasets, there are also efforts underway to create corpora for languages which have limited authentic written data. For example, Doumbouya et al. (2021) collected voice data (from radio broadcasts) to create voice recognition systems for West African languages. [Seamless4MT](#) from Meta provides speech-to-speech translations, as well as speech-to-text, text-to-speech, and text-to-text for up to 100 languages (Singh et al., 2024). [NusaCrowd](#) uses multimodal input to create a standardized collection of multiple data sets covering 19 local Indonesian languages (Cahyawijaya et al., 2023). These projects are often underfunded, with work done by volunteers and nonprofits in spite of difficult technical and practical issues (Bird, 2020). Cooperation with local universities is often a vital part of such projects, as they are able to contribute both funding and expertise (Elliott, 2021a; Wiecheteck et al., 2024).

The integration of minority languages into non-commercial and locally produced LLMs is an important step towards normalization for the users, learners, and teachers of those languages. However, such systems are available for relatively few languages and are mostly in the development or prototype stage. For the foreseeable future, the mainstream AI systems will continue to be the go-to AI solutions, *ChatGPT* in particular. That means that the cultural and linguistic biases will remain. It is also the case that for all languages, AI systems will persist to exhibit deficiencies in dealing with nuanced social and cultural contexts (Godwin-Jones, 2024b). Users of all kinds should be aware of the limitations and approach the use of AI with skepticism from a perspective of critical AI literacy (Darvin, 2023b; Godwin-Jones, 2024a).

The limitations of AI for robust performance using minority languages needs to be recognized but should not be an impediment to experimenting with AI to test its effectiveness in a given language (Dizon, 2024; Li et al., 2023). The learning and teaching opportunities are too great to ignore. They include, for example: (a) to use AI to create appropriate and contextually tailored learning materials; (b) to provide out-of-class writing or speaking practice in a setting that conforms to the learner’s preferred time and place; and (c) to provide assessment of language use and provide custom feedback. Song and Tang (2025, this issue) demonstrated how (Chinese) learning can be improved through the personalized feedback available from AI systems, while Shin and Choi (2025, this issue) showed how AI chatbots improve

learners' (Korean) lexico-grammar, especially for lower proficiency learners. AI-powered machine translation is shown to be effective when integrated into plurilinguistic learning (Olthuis & Gerstenberger, 2019; van Lieshout & Cardoso, 2022; Vogel et al., 2018). As AI increasingly incorporates multimodal data and gradually expands the list of supported languages, the situation for minority languages is likely to improve. AI may or may not be a game changer for minority languages, but for everyone it will likely be a phenomenon that cannot be pushed aside, given its outsized role in all aspects of society.

Conclusion

Technology is integrated into the teaching of LCTLs considerably more now than was the case in 2013, the year of the last *LLT* issue on minority languages. In part, that has been driven by necessity, as the COVID pandemic led to widespread use of online learning. Online classes and social media allow geographically separated LX learner-users to become communities of practice. At the same time, there has been an uptick in the creation of dedicated corpora and digital archives for ever more languages. Fortunately, those resources tend to be freely shared, so as to be available for both organized learning and individual self-study. Local grassroots organizations and scholars have often been in the forefront of collecting and creating learning resources, an effort unevenly supported and funded by government agencies.

Generalizability across languages in terms of technology resources for language learning is not possible, as conditions vary tremendously. Resources tend to be most abundant for languages officially recognized (Irish, Icelandic; O'Toole et al., 2023) or that have seen successful revitalization efforts (Hawaiian or Māori; Elliott, 2021b). Indigenous languages, many of which are endangered, offer a special case, while also varying widely in terms of number of speakers, relationship to majority languages, learning opportunities, and available support structures. The degree of precarity of a language, along with the attitudes of local communities, are likely to contribute substantially to determining the availability of online and digital resources. As discussed above, the intimate connection of language, culture, and place may lead in some cases to a wariness in accepting the viability of online learning, disconnected as it is physically from communities and their environments. Those concerns can also impact teaching formats and strategies, leading towards or away from pedagogical translanguaging and the integration of AI.

Concerns over exploitation and a loss of control may result in an insistence on data sovereignty for Indigenous communities (Hao, 2022; McKnight & Shipp, 2024). Those concerns may justifiably be extended to generative AI. For both projects in AI and in pedagogical translanguaging, it will be vital for projects to not just rely on outside experts, but to make sure to integrate community members so that projects are sustainable over the long run (Byrnes, 2020). For Indigenous languages, it is important not only to safeguard appropriately linguistic resources, but also to be even more cautious and critical in integrating AI. That critical consciousness is ideally also incorporated into the practice of pedagogical translanguaging, so that students not only build their custom repertoire of plurilingual semiotic resources, but also gain insights into how standard national languages represent power and prestige that need to be recognized and taken into account in language practices in social, educational and professional settings (García & Otheguy, 2019). That can lead to insights into how social justice in many contexts is connected to the mix of languages used in a given context and to the social hierarchy of local and majority languages (Alberto Mora et al., 2022; Canagarajah, 2021; Meighan, 2021).

The current evolution of technology favors increasing accessibility to language resources and to AI systems anytime and anywhere the user might be (Godwin-Jones, 2021). Mobile devices serve as effective mechanisms for integrating plurilingualism and AI-powered resources, combining multilingual social media, streaming services, chat interfaces, and language tools (translation apps, dictionaries, phrase books). That powerful combination has been shown to be an invaluable resource for migrants on the road (Yilmaz, 2024), for marginalized Indigenous communities (Oliver & Nguyen, 2017), and for immigrants finding their way into new linguistic and cultural settings (Kukulska-Hulme, 2019). That dynamic points to the increasing presence in our world of "human-machine assemblages" (Vogel et al., 2018, p. 90)

represented by the smartphone and emerging connected wearables, as meaning-making resources expand to multimodalities, multiple languages, and AI-powered apps (Godwin-Jones, 2017). In that process, AI joins with mobile technologies to represent a new kind of joint human-machine agency, as well as serving as a potentially powerful learning and teaching partner (Godwin-Jones, 2024a). Digital devices function as “cognitive prostheses” (Tzirides, 2024, p. 101) in that they extend our senses (camera, sensors, microphones) and our memories.

That prospect of digital empowerment is not universal; technology resources the CALL community sometimes assumes as givens are anything but that in underserved and disadvantaged communities (Godwin-Jones, 2024a; McKnight & Shipp, 2024). At the same time, critics have raised concerns over the ethics and equity of AI use (Bender et al., 2021) and pedagogical translanguaging (Mendoza et al., 2024). Sociopolitical realities impinge on all technologies and pedagogies, so that we always need to be alert to existing power hierarchies and entrenched interests and make sure a critical, balanced assessments are passed on to our learners when we advocate for change in education (Freire, 2000/1968). Needless to say, that is all the more strongly the case for Indigenous communities and cultures, whose interests and voices/languages evince a long history of neglect and existential threat.

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